

Logistic Regression: Foundations and Application

Statistics for Social Work

Models for Binary Outcomes

Why Logistic Regression?

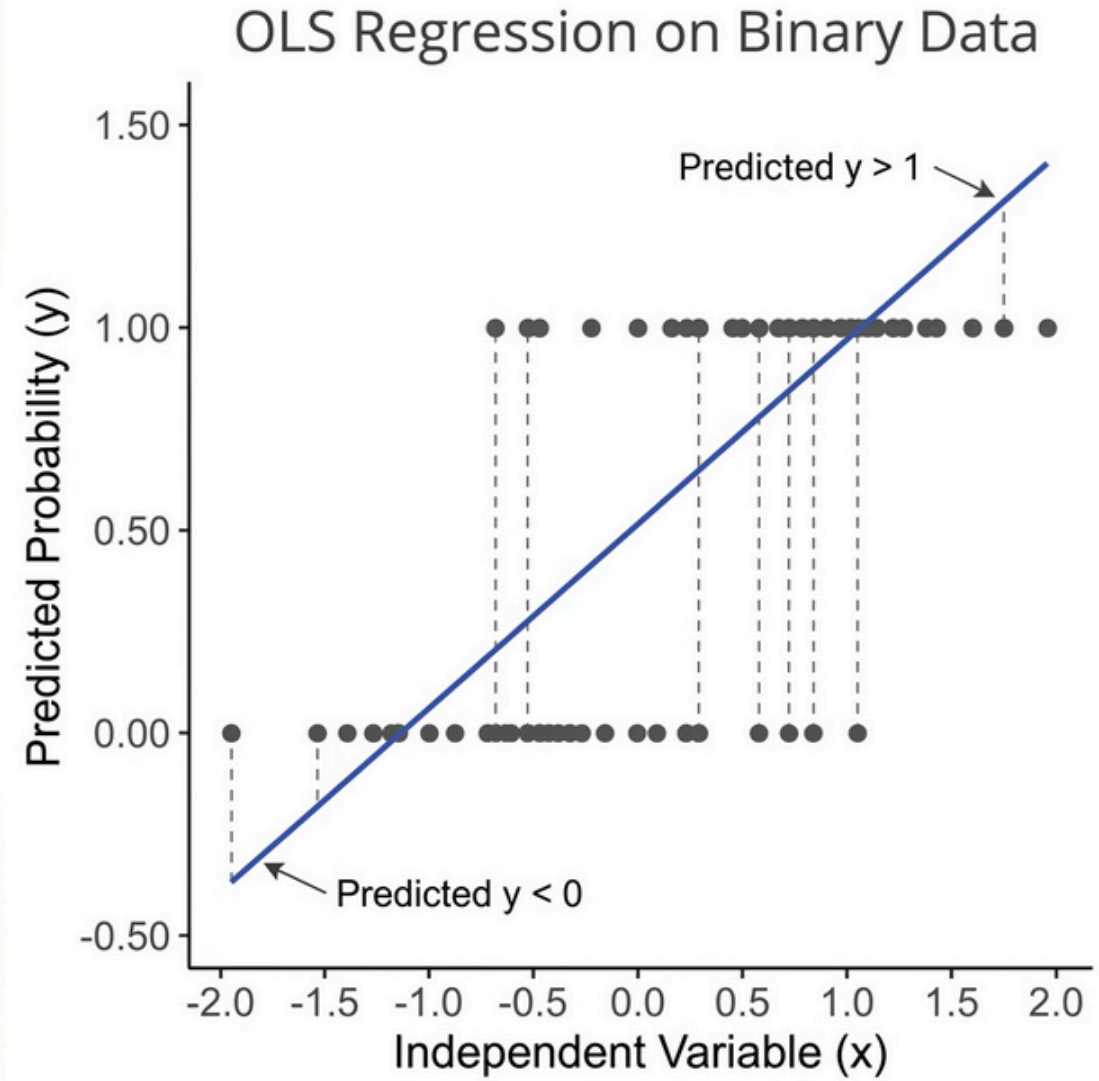
In social work research, we often examine **binary outcomes**:

- Does a child re-enter foster care? (Yes/No)
- Does a client complete a substance abuse program? (Success/Failure)
- Is a housing policy effective in preventing eviction? (Evicted/Not Evicted)

The Problem with OLS

When we try to use Ordinary Least Squares (OLS) for these outcomes, we run into major issues:

- **Error terms** are not normally distributed.
- **Homoscedasticity** is violated.
- **Probabilities** can be predicted as < 0 or > 1 (Impossible!)



The Solution: Generalized Linear Models (GLMs)

When OLS fails, we move into the world of **GLMs**.

Think of a GLM as a flexible "umbrella" that allows us to model data that:

- **Isn't normal** (e.g., Yes/No outcomes).
- **Isn't continuous** (e.g., counts of incidents).
- **Is bounded** (e.g., probabilities must stay between 0 and 1).

In social work, our most interesting questions usually involve these "messy" outcomes.

The Three Core Components of a GLM

To build any GLM, you need three "ingredients":

1. The Random Component (The Distribution):

What "shape" does your outcome take?

- Normal (OLS)
- **Binomial** (Logistic Regression)
- Poisson (Count models)

The Three Core Components (Cont.)

2. The Systematic Component (The Predictors):

The familiar linear part: $\beta_0 + \beta_1 X_1 + \dots$

- This part looks exactly like OLS!

3. The Link Function (The Bridge):

The mathematical tool that "stretches" or "shrinks" our predictors to match the outcome.

- It connects the mean of the outcome to our linear predictors.

Why Logistic Regression is a GLM

Logistic Regression is just a specific type of GLM where:

1. **Random:** The outcome follows a **Binomial distribution** (0 or 1).
2. **Systematic:** We use linear predictors (β).
3. **Link:** We use the **Logit Link** to bridge the two.

Theoretical Goal: We use the Logit to transform probabilities (0 to 1) into a linear scale ($-\infty$ to $+\infty$) that our model can handle.

From Probability to Odds

Before we reach the model, we must understand the language of chance.

Probability (P):

The likelihood of an event occurring, bounded between 0 and 1.

$$P = \frac{\text{Events}}{\text{Total Observations}}$$

Odds:

The ratio of the probability of the event occurring to the probability of it *not* occurring.

$$\text{Odds} = \frac{P}{1-P}$$

Practice: Calculating Odds

Imagine a study of 100 families in a home-visiting program.

- 20 families experienced a case closure due to non-compliance ($P = 0.20$).
- 80 families remained compliant ($1 - P = 0.80$).

Odds of Non-compliance:

$$\text{Odds} = \frac{0.20}{0.80} = 0.25$$

Interpretation: For every 1 family that is non-compliant, there are 4 compliant families.

The Logit Link: Log-Odds

The "Logistic" in Logistic Regression refers to the **Logit Link Function**.

Odds are asymmetrical (0 to ∞). To make them "linear" and symmetrical ($-\infty$ to $+\infty$), we take the natural logarithm.

$$\text{Logit}(P) = \ln(\text{Odds}) = \ln\left(\frac{P}{1-P}\right)$$

When $P = 0.5$, Odds = 1, and $\ln(1) = 0$.

The Logistic Regression Equation

Now we can form a linear expression similar to OLS, but for the Logit:

$$\ln \left(\frac{P}{1 - P} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

- β_0 : The intercept (log-odds when all predictors are zero).
- β_1 : The change in **log-odds** for a one-unit increase in X_1 .

Wait—log-odds are hard to explain to stakeholders! We need to go back.

Returning to Odds Ratios (OR)

To make the results "interpretable," we exponentiate the β coefficients.

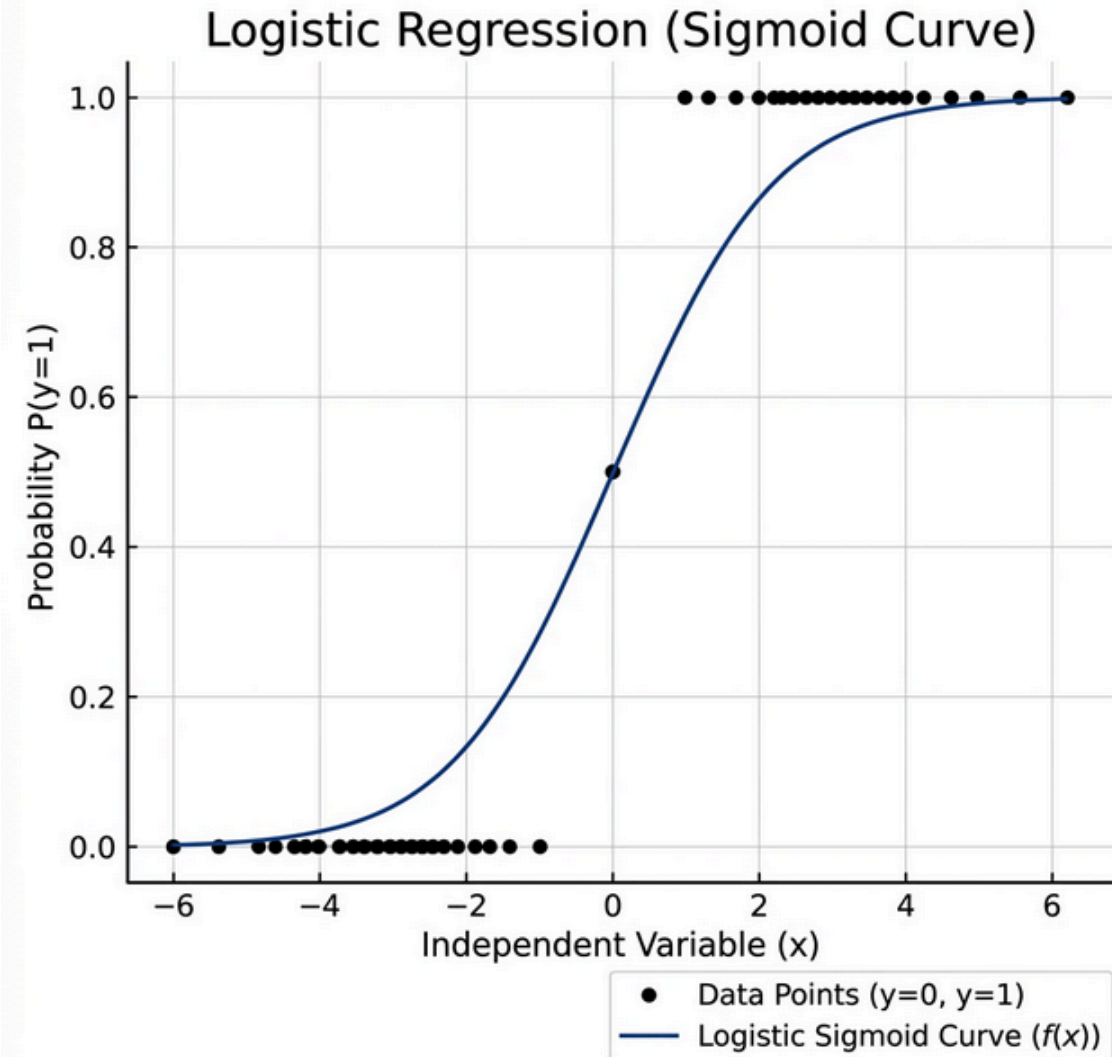
$$\text{Odds Ratio (OR)} = e^{\beta_1}$$

- **OR > 1**: The event is **more** likely as X increases.
- **OR < 1**: The event is **less** likely as X increases.
- **OR = 1**: No relationship.

Theoretical: The Exponential Curve

In OLS, we have a straight line. In Logistic Regression, the relationship between X and **Probability (P)** is an **S-shaped curve** (Sigmoid).

- At the extremes, the curve flattens.
- This represents "diminishing returns" on probability change as you move away from $P = 0.5$.
- Small changes in X have the biggest impact on P when the probability is near 50%.



Predicting Probability

To get the actual probability (P) for a specific case:

$$P = \frac{e^Z}{1 + e^Z}$$

Where $Z = \beta_0 + \beta_1 X_1 + \dots$ (the predicted logit).

Or, rearranged:

$$P = \frac{1}{1 + e^{-Z}}$$

Assumption 1: Binary Response

The dependent variable must be truly dichotomous.

- Yes/No, Living/Dead, Graduated/Dropped Out.
- If you have 3+ categories, you need **Multinomial Logistic Regression**.
- If categories are ranked, you need **Ordinal Logistic Regression**.

Assumption 2: Linearity of Predictors

Crucial Distinction:

Logistic regression does not assume X is linearly related to P .

It assumes X is linearly related to the **Log-Odds (Logit)**.

- If this is violated, your coefficients are biased.
- Tested using the Box-Tidwell transform or looking at residuals.

Assumptions: Independence & Sample Size

- **Independence of Observations:** The "No Clustering" rule. Observations should not be related (e.g., students in the same school). If clustered, use **Multilevel Models**.
- **Multicollinearity:** Predictors should not be too highly correlated ($r > 0.8$).
- **Sample Size:** Need at least 10-15 "Events Per Variable".

Part 2: Practical Mechanics

Predicting Foster Care Re-entry

Let's imagine a dataset where:

- **Outcome (Y):** Re-entry into Foster Care within 12 months (1 = Yes, 0 = No).
- **Predictors:**
 - i. X_1 : Number of previous placements (Continuous).
 - ii. X_2 : Parental visitation frequency (Scale 1-5, 5 = High).

The Model Output (Dummy Data)

Variable	Coefficient (β)	Std. Error	P-value	Exp(β) [OR]
Intercept	-2.50	0.40	<0.001	0.082
Prev Placements	0.45	0.15	0.003	1.568
Visitation	-0.60	0.20	0.002	0.549

Interpreting "Previous Placements"

$$\beta_{Placements} = 0.45, OR = 1.57$$

For every additional previous placement:

- The **log-odds** of re-entry increase by 0.45.
- The **odds** of re-entry increase by **57%** ($1.57 - 1.00 = 0.57$).
- *Social Work Context:* Higher instability in early system contact predicts future instability.

Interpreting "Visitation"

$$\beta_{Visitation} = -0.60, OR = 0.55$$

For every one-unit increase in visitation frequency:

- The **log-odds** of re-entry decrease by 0.60.
- The **odds** of re-entry are **45% lower** ($1.00 - 0.55 = 0.45$).
- *Social Work Context:* Maintaining the parent-child bond is a strong protective factor against re-entry.

Calculating a Specific Case

Client: Maria

- Previous Placements = 3
- Visitation Score = 2

Step 1: Calculate the Logit (Z)

$$Z = -2.50 + (0.45 \times 3) + (-0.60 \times 2)$$

$$Z = -2.50 + 1.35 - 1.20 = -\mathbf{2.35}$$

Calculating Maria's Probability

Step 2: Convert Logit to Probability

$$P = \frac{e^{-2.35}}{1 + e^{-2.35}}$$

$$e^{-2.35} \approx 0.095$$

$$P = \frac{0.095}{1 + 0.095} = \frac{0.095}{1.095} \approx \mathbf{0.087}$$

Maria has an 8.7% predicted probability of re-entry.

What if Case Complexity Increases?

Client: James

- Previous Placements = 6
- Visitation Score = 1

Logit (Z):

$$Z = -2.50 + (0.45 \times 6) + (-0.60 \times 1)$$

$$Z = -2.50 + 2.70 - 0.60 = -\mathbf{0.40}$$

Probability (P):

$$P = \frac{e^{-0.40}}{1+e^{-0.40}} = \frac{0.67}{1.67} \approx \mathbf{0.401}$$

James has a 40.1% predicted probability of re-entry.

Evaluating the Model: Fit Measures

We don't have R^2 in logistic regression. Instead, we use:

1. **Pseudo R^2 (Cox & Snell, Nagelkerke):** Estimates the proportion of variance explained (usually lower than OLS R^2).
2. **Log-Likelihood:** How "likely" the observed data is given the model parameters.
3. **Wald Test:** Tests if a specific β is significantly different from zero (similar to the t -test in OLS).

The Confusion Matrix

A tool for classification accuracy.

	Predicted: No	Predicted: Yes
Actual: No	True Negatives (TN)	False Positives (FP)
Actual: Yes	False Negatives (FN)	True Positives (TP)

- **Sensitivity:** Ability to correctly identify those who *do* re-enter.
- **Specificity:** Ability to correctly identify those who *do not*.

Numerical Example: 100 Children

Let's see the metrics in action for our Foster Care model.

	Predicted: NO	Predicted: YES	Total
Actual: NO	63 (TN)	7 (FP)	70
Actual: YES	9 (FN)	21 (TP)	30
Total	72	28	100

Calculations:

- **Sensitivity:** $21/30 = 0.70$ (70%)
- **Specificity:** $63/70 = 0.90$ (90%)
- **False Positive Rate:** $7/70 = 0.10$ (10%)
- **False Negative Rate:** $9/30 = 0.30$ (30%)

Interpreting the Errors

In social work, errors have human consequences.

1. False Positive (Type I Error): 7 children

- Model said they would re-enter, but they didn't.
- *Consequence:* Families may be subjected to unnecessary, intrusive surveillance or services.

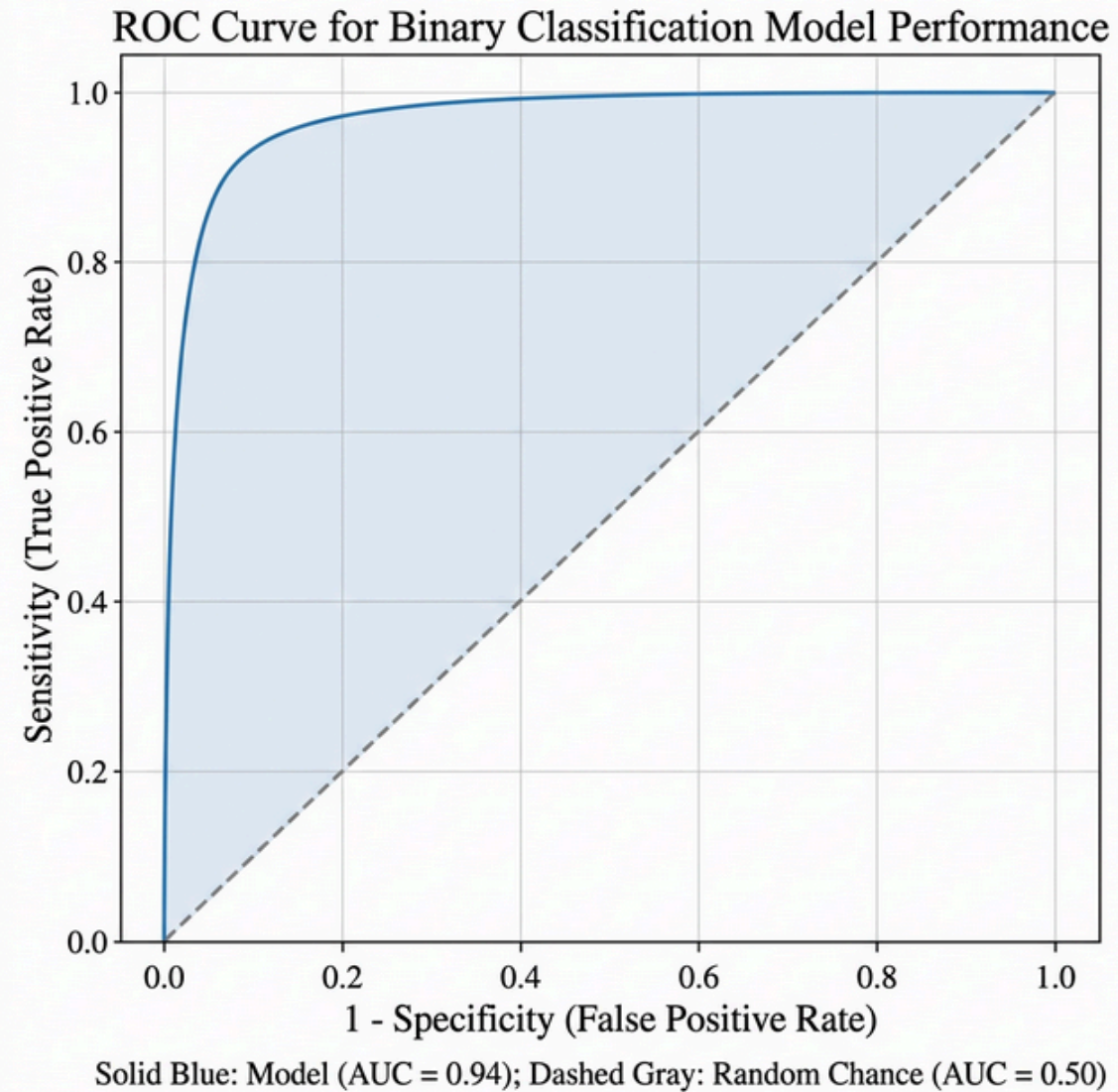
2. False Negative (Type II Error): 9 children

- Model said they were safe, but they re-entered.
- *Consequence:* Systemic failure to protect. The child returns to an unsafe environment.

Which error is more "costly"? This is a policy decision, not a math one.

The Sensitivity Plot (ROC Curve)

The metrics change as we move our "cutoff"
(e.g., from 0.5 to 0.7).



Understanding the ROC Curve

Receiver Operating Characteristic (ROC):

- **X-axis:** 1 - Specificity (False Positive Rate).
- **Y-axis:** Sensitivity (True Positive Rate).

Interpretation:

- **The Diagonal Line:** Represents a model that is "guessing" (Random Chance).
- **The Curve:** Represents our model's performance across all possible thresholds.
- **Area Under the Curve (AUC):**
 - **0.5:** No better than a coin flip.
 - **0.7 - 0.8:** Acceptable classification.
 - **0.8 - 0.9:** Excellent.
 - **>0.9:** Outstanding (but check for overfitting!).

Model Fit: The Deviance Statistic

In Logistic Regression, we use **Maximum Likelihood Estimation (MLE)**.

Likelihood (L): The probability of seeing the observed data given our model parameters.

-2 Log Likelihood (-2LL):

- Also called the **Deviance**.
- It is a measure of "badness of fit"—how much the model *deviates* from the data.
- **Goal**: Minimize the Deviance. A perfect model has a Deviance of 0.

The Likelihood Ratio Test

How do we know if our predictors actually help? We compare models using the **Likelihood Ratio (LR) Test**.

Comparison:

- **Null Model:** Intercept only (no predictors).
- **Proposed Model:** Our model with predictors.

$$G^2 = \text{Deviance}_{\text{Null}} - \text{Deviance}_{\text{Proposed}}$$

The difference (Δ Deviance) follows a **Chi-Square (χ^2) distribution**. If the change is significant, our model is better than chance.

Practical Tips for Your Dissertation

1. **Always report OR and 95% Confidence Intervals.**
2. **Check your EPV.** Don't over-model a small sample of events.
3. **Consider the "Baseline Odds."** If re-entry is very rare ($< 5\%$), the model behaves differently than if it is 50/50.
4. **Think about Interactions.** Does visitation matter *more* for children with fewer placements? (Add an interaction term: $X_1 \times X_2$).

Part 3: Real-World Practice (JASP)

Case 1: Predicting College Enrollment

Let's look at our first dataset:

- **Outcome (Y):** College Enrollment (1 = Enrolled, 0 = Not).
- **Predictor (X):** Undergraduate GPA.

The Question: How much does GPA improve your odds of enrolling?

The Coefficients Table (JASP)

Predictor	β	Odds Ratio [OR]	P-value
Intercept	-2.144	0.117	0.375
GPA	0.883	2.419	0.282

Interpretation:

- The equation is:

$$\text{logit}(Y) = -2.144 + 0.883(GPA)$$
- For every 1-unit increase in GPA, the **odds** of enrolling are **2.42 times greater** ($e^{0.883} = 2.419$).

Focus on the results first

Coefficients

Model		Estimate	Standard Error	Odds Ratio	z	Wald Test	
						Wald Statistic	p
M ₀	(Intercept)	0.405	0.645	1.500	0.628	0.395	1
M ₁	(Intercept)	-2.144	2.416	0.117	-0.888	0.788	1
	gpa	0.883	0.821	2.419	1.076	1.157	1

Note. enroll level '1' coded as class 1.

$$\text{logit}(Y) = -2.144 + .833GPA$$

$$p(X) = \Pr[Y = 1|X = x] = \frac{\exp(-2.144 + .883X)}{1 + \exp(-2.144 + .883X)}$$

Probability of enrolling in college 'conditional on' values of one's grade point average

- The Wald test is the test statistic for the logistic regression model
- The sig. is the p-value
- EXP(B) is the odds ratio – the interpretation is: the odds of enrolling in college are about 2.42 times greater for each one unit increase in undergraduate gpa, but the result is not statistically significant

Computing Probabilities

- The most useful part of logistic regression is the ability to predict the conditional probability of Y given values of the independent variables
- For example: what is the probability of college enrollment given that your GPA is 1.0, 2.0, 3.0 and 4.0?

$$Pr(Y = 1) = p = \frac{e^{\beta_0 + \beta_1 X_1}}{1 + e^{\beta_0 + \beta_1 X_1}}$$

Examples:

- $Pr(Y = 1|X = 1) = \frac{e^{-2.144 + .883(1)}}{1 + e^{-2.144 + .883(1)}} = \frac{e^{-1.261}}{1 + e^{-1.261}} = \frac{.283371}{1.283371} = \mathbf{.221}$
- $Pr(Y = 1|X = 2) = \frac{e^{-2.144 + .883(2)}}{1 + e^{-2.144 + .883(2)}} = \frac{e^{-.378}}{1 + e^{-.378}} = \frac{.685}{1.685} = \mathbf{.407}$

Computing Probabilities (Step-by-Step)

We convert the logit back to a probability using: $P = \frac{e^{\text{logit}}}{1+e^{\text{logit}}}$

Example Calculation for GPA = 3.0:

1. **Find Logit:** $-2.144 + 0.883(3.0) = 0.505$

2. **Exponentiate:** $e^{0.505} = 1.657$

3. **Calculate P:** $\frac{1.657}{1+1.657} = 0.623$

Result: A student with a 3.0 GPA has a **62.3% probability** of enrolling.

Table of Predicted Probabilities

GPA (X)	β_0	β_1	e^{logit}	$1 + e^{\text{logit}}$	Prob (P)
1	-2.144	0.883	0.283	1.283	0.221
2	-2.144	0.883	0.685	1.685	0.407
3	-2.144	0.883	1.657	2.657	0.624
4	-2.144	0.883	4.007	5.007	0.800

Notice how the relationship is not a straight line! Moving from a 1.0 to 2.0 has a different impact than moving from 3.0 to 4.0.

Improving Classification

By default, we classify someone as "Enrolled" if their $P > 0.50$.

Model Summary & Classification

- The classification table is provided. Here we correctly classify 60% of enrollments, better than chance (not much better)
 - Using the 50% cutoff, when the observed Y was the student did not enroll in college, we predicted enrollment correctly 1/2 of the time
 - Similarly, using the same cutoff, when the student did enroll, we were correct 4 of 6 times (2/3)

Confusion matrix

Observed	Predicted		% Correct
	0	1	
0	2	2	50.000
1	2	4	66.667
Overall % Correct			60.000

Note. The cut-off value is set to 0.5

Specificity
Sensitivity

$$(2 + 4) / (2 + 2 + 2 + 4) = 60\%$$

Improving Classification (Cont.)

What if we change the "Cutoff" to 0.40?

- We become more "liberal" in predicting enrollment.
- Some people previously predicted as "Not Enrolled" switch to "Enrolled".

Result: In this specific dataset, our accuracy jumped from 60% to 80%. *But be careful—this can increase your "False Positives"!*

Classification Table^a

Observed		Predicted		Percentage Correct
		enroll	enroll	
Step 1	enroll	.00	2	50.0
		1.00	0	100.0
Overall Percentage				80.0

a. The cut value is .400

Default = .5			Default = .4		
gpa	gender	enroll	PRE_1	PGR_1	PGR_2
1.89	.00	.00	.38350	.00	.38350
1.70	.00	.00	.34467	.00	.34467
4.15	.00	1.00	.82078	1.00	.82078
2.72	1.00	1.00	.56425	1.00	.56425
3.16	1.00	.00	.65636	1.00	.65636
3.89	1.00	1.00	.78448	1.00	.78448
4.02	1.00	1.00	.80326	1.00	.80326
2.10	.00	1.00	.42819	.00	.42819
2.36	.00	1.00	.48511	.00	.48511
3.55	1.00	.00	.72941	1.00	.72941

When we changed the default value from .5 to .4, what changed?

Better Than Chance? (Press's Q)

Is 60% accuracy actually good? For a 50/50 outcome, maybe not.

You can report **Press's Q**, which is distributed as χ^2 :

$$Q = \frac{[N - (nK)]^2}{N(K - 1)}$$

- N : total sample size (10)
- n : number of cases correctly classified (6)
- K : number of groups (2)

$$Q = \frac{[10 - (6 \times 2)]^2}{10(2 - 1)} = 0.4$$

Since $0.4 < \chi^2(1) = 3.841$, the results are **no better than chance**.

Model Comparison: Does Gender Help?

In social science, we often ask: "Does adding more information make our model better?"

Model 0: GPA only.

Model 1: GPA + Gender.

The Verdict:

- The p -value for the block change (0.256) is greater than 0.05.
- Adding "Gender" **did not significantly improve** our ability to predict college enrollment. *Keep it simple—stick to GPA!*

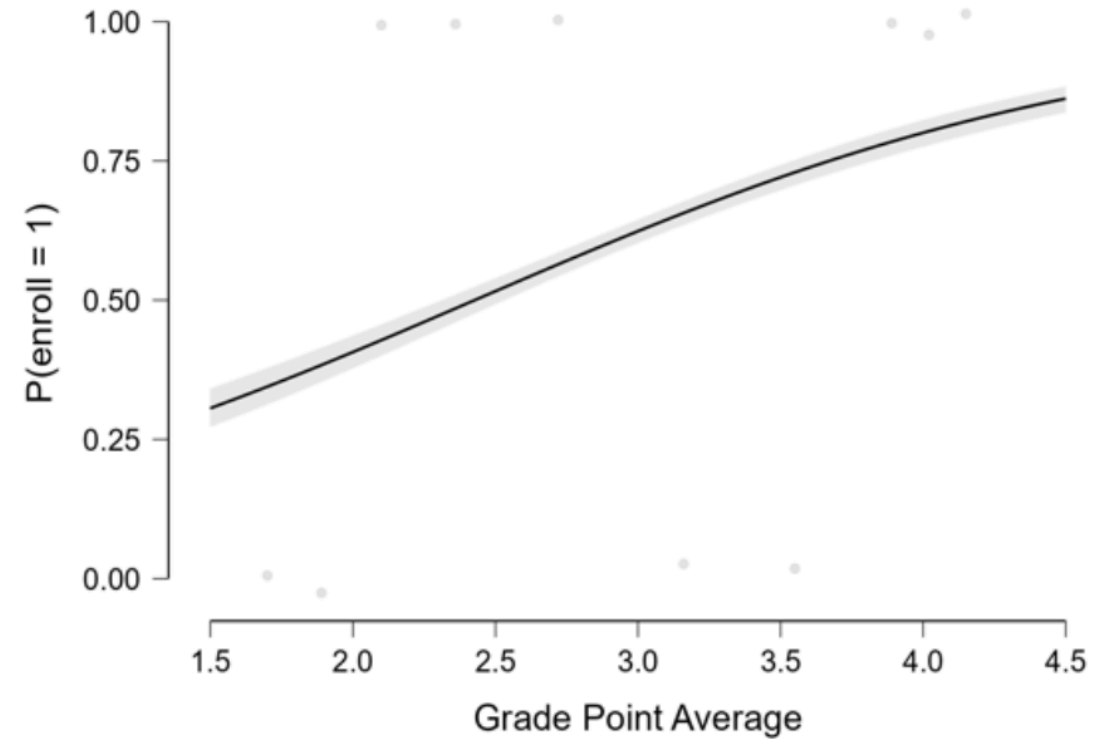
Model Summary - enroll

Model	Deviance	AIC	BIC	df	ΔX^2	p	McFadden R ²	Nagelkerke R ²	Tjur R ²	Cox & Snell R ²
M ₀	13.460	15.460	15.763	9			0.000		0.000	
M ₁	12.169	16.169	16.775	8	1.291	.256	0.096	0.164	0.119	0.121

Note. M₁ includes gpa

Visualizing the Probability Curve

Logistic regression doesn't predict "Yes/No" directly—it predicts the **Probability** (P).



Visualizing the Probability Curve (Cont.)

- **The Shape:** Notice the "S-curve" (Sigmoid).
- **The Mid-Point:** Where $P = 0.50$ (around GPA 2.4).
- **The "Floor" and "Ceiling":** No matter how low or high the GPA, the probability never drops below 0 or goes above 1.

This is why OLS failed us! OLS would predict probabilities like -0.2 or 1.5.

Practice Example: Passing Statistics

Scenario: Predict passing a stats test (1 = Pass, 0 = Fail) based on hours of study.

- Open the SPSS file passing.sav
- The data has three made-up variables: Passed a statistics test; hours of study, gender (1 = male, 0 = female)
- Run a binary logistic regression of passed on hours
- Write out the regression equation
- Predict the probability of passing for 0, 50, 100, 150, 200 and 250 hours of studying
- Add gender into the model and interpret the coefficient in terms of odds and percent change in odds
- Does adding gender improve the model fit?

Model & Interpretation

$$\text{logit}(Y) = -9.3 + .082X$$

$$p(x) = P[Y = 1|X = x] = \frac{\exp(-9.3 + 0.082X)}{1 + \exp(-9.3 + 0.082X)}$$

Table 1. Predicted probability of Passing Conditional on Hours Studying

X	0	50	100	150	200	250
$P(Y x = X)$	0.0055	0.0603	0.4286	0.6298	0.8975	0.9783

- Each additional hour spent working increases the odds of passing by $[\exp(.082) - 1 \times 100]$ percent
- Each additional hour spent working increases the odds of passing by about 8.5 percent
- The odds of passing are 1.085 times higher for each additional hour spent studying

Case 2: Adding Gender

- **Equation:**

$$\text{logit}(\text{Passing}) = -8.924 + 0.079(\text{Hours})$$

- **Gender (1 = Male) OR = 0.520.**

- **Interpretation:** In this sample, being Male is associated with **48% lower odds** of passing compared to females ($0.520 - 1 = -0.48$).

- *Note:* This was not a statistically significant improvement ($p = .862$).

Let's add a binary variable, gender, to the equation

Coefficients

	Estimate	Standard Error	Odds Ratio	z	Wald Test		
					Wald Statistic	df	p
(Intercept)	-8.924	8.487	1.332e-4	-1.051	1.106	1	0.293
hours	0.079	0.071	1.083	1.126	1.267	1	0.260
gender (1)	-0.654	3.764	0.520	-0.174	0.030	1	0.862

Note. passed level '1' coded as class 1.

The odds of passing for males are $[\exp(-.654)-1 * 100]$ percent lower compared to females

The odds of passing for males are 48.01% lower compared to females

The odds of passing are .5100 ($\exp(-.654)$) times lower for males compared to females

The file probability of passing by gender.xlsx has the probability of passing for males and females given 100 hours of study time

Summary

Logistic Regression is the bridge between qualitative social work outcomes and quantitative prediction.

- **Outcome:** Binary (1/0).
- **Link:** Logit (Natural log of odds).
- **Interpretation:** Odds Ratios are your best friend for communication.
- **Goal:** Predict the probability of life events to better target interventions.

Q&A

Discussion points:

1. What are ways that programs and social service providers can reduce food insecurity among adults experiencing homelessness?
2. What are ways that programs and social service providers can reduce poor health, risky health behaviors, and the trauma associated with experiencing personal or sexual victimization among adults experiencing homelessness?